Campus Lost & Found Management System

Problem Statement and Objectives

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Course: DBMS and OOPS

Problem Statement

Many campuses still manage lost and found items through notice boards, word of mouth or paper registers. These methods are slow and unreliable. Students cannot easily check whether a lost item has been found, records are easily misplaced, there is no standard way to verify ownership, and administrators cannot track what happened to each item.

Existing System Limitations

1. There is no central record of lost and found items.
2. Items cannot be searched by keyword or status.
3. There is no proper claim process to verify ownership.
4. There is no tracking of whether an item is lost, found or claimed.
5. Manual records are easily lost, damaged or duplicated.

Proposed System

The Campus Lost & Found Management System is a Java desktop application with a Swing interface, built on object-oriented principles and connected to a MySQL database through JDBC. Students can report lost or found items, search for them and submit claims. Administrators review each claim and approve or reject it, and the item status is updated automatically after an approved claim.

Users

1. **Students:** report items, search for items, submit claims and view claim status.
2. **Administrators:** manage items and approve or reject claims.

Objectives (Functional and Design)

Functional Objectives

1. Store all lost and found item records in a central MySQL database.
2. Provide add, view, update and delete operations for items, with input validation.
3. Provide keyword search and status filters to find items quickly.
4. Allow students to submit claims and view their claim status.
5. Allow administrators to approve or reject claims, and update the item status after a successful claim.

Design Objectives

6. Apply the OOP concepts of encapsulation, inheritance, polymorphism and abstraction, using an abstract *Item* class extended by *LostItem* and *FoundItem*.
7. Separate the user interface, service and DAO layers, so that no SQL queries are written inside Swing forms.
8. Connect Java to MySQL using JDBC with a reusable database connection class.

Scope

The system is a standalone desktop application for use within a single campus. It covers item management, search, and the claim and approval workflow. Online or mobile access, email notifications and image-based item matching are outside the scope of this project.